

Cataract Phacoemulsification Surgery

Beginner-to-Advanced Step-by-Step Operative Field Guide & Anatomical Rationale

TARGET PATHOLOGY
Nuclear Cataract (LOCS III
NO1-NO5)

PRIMARY INCISION
2.4 mm Tri-Planar Clear
Cornea

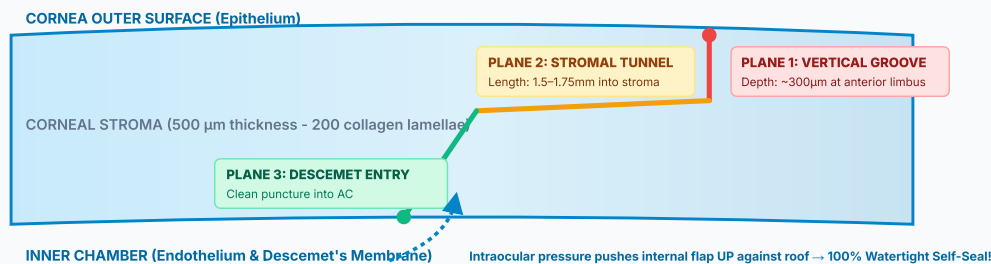
ULTRASOUND FREQUENCY
28–45 kHz Piezoelectric

ANESTHESIA PROTOCOL
Topical Proparacaine +
Intracameral Lidocaine

☀ What Is Cataract Surgery in Simple Terms?

Inside your eye is a natural crystal lens that focuses light onto your retina. As you get older, the proteins inside clump up and turn cloudy like frosted bathroom glass—this is a **cataract**. During surgery, we make a tiny self-sealing doorway into the eye, cut a round window in the cellophane-like bag holding the cataract, use a microscopic ultrasonic jackhammer (phacoemulsification) to turn the hard rock into soup and vacuum it out, and leave the clean empty bag ready for a new artificial lens.

CORNEAL INCISION ARCHITECTURE & TRI-PLANAR SELF-SEALING PHYSICS



STEP 1 Paracentesis Incision (~1.0mm MVR) — The Small Side Door

Tool: 1.0mm MVR Blade

☞ WHAT TO DO RIGHT NOW:

Pick up the 1.0mm MVR blade from your tray. Locate the 10 o'clock position (upper left) on the edge of the clear cornea. Angle the blade flat (parallel to the iris plane) and make a single, crisp 1-millimeter puncture into the front chamber.

💡 **Why This Step Is Strictly Necessary:** Eye surgery is a two-handed operation. Your primary hand operates the phaco ultrasound needle through the main wound, but you need a second doorway for an assistant tool (such as a Nagahara chopper, cyclodialysis spatula, or nucleus rotator) in your non-dominant hand. The second instrument holds the slippery cataract chunks in place, rotates the lens, and protects the back of the eye. Without this side port, you would have zero control over the cataract chunks inside the eye.

Anatomical Target	Peripheral clear cornea at the 10 o'clock limbus, just anterior to the fine red blood vessels (vascular arcade).
Surgical Technique	Keep blade strictly parallel to the colored iris. Avoid aiming downward (which would puncture the iris or tear the front lens capsule) or aiming too upward (which tears the corneal stroma).

⚠ **Critical Hazard:** Entering too anteriorly in the clear cornea induces corneal astigmatism and distortion. A sudden downward plunge will lacerate the iris and induce massive bleeding (hyphema).

👉 **WHAT TO DO RIGHT NOW:**

Switch to the 2.4mm angled keratome blade. Position the blade at 1:30 o'clock on the limbus. Cut in 3 continuous steps: 1) Make a vertical 300µm groove; 2) Tunnel forward 1.5mm through the cornea wall; 3) Dimple down to enter the eye chamber.

💡 **Why This Step Is Strictly Necessary:** The phacoemulsification needle is 2.4mm wide. You must create an entry path that fits the instrument snuggly without leaking irrigation fluid. Cutting it in 3 staggered planes (tri-planar) creates an automatic self-sealing one-way valve: natural pressure from inside the eye pushes the internal corneal flap against the roof of the tunnel, sealing it shut 100% watertight without requiring stitches or sutures.

Plane 1 (Vertical Groove)	Perpendicular vertical groove (~300 µm depth, half corneal thickness) at the anterior limbus.
Plane 2 (Lamellar Tunnel)	Blade flattened to advance 1.5–1.75 mm through the dense collagen fibers toward the corneal center.
Plane 3 (Descemet Entry)	Tip dimpled downward to cleanly pierce Descemet's membrane with a crisp internal entry.

⚠️ **Critical Hazard:** A tunnel that is too short (<1.2 mm) will leak, causing the eye to go soft, iris prolapse, and catastrophic post-op infection (endophthalmitis). A tunnel that is too long (>2.2 mm) creates corneal wrinkles (striae) and blocks tool movement.

STEP 3

OVD Injection (Dispersive & Cohesive) — The Protective Jelly Shield

Tool: Viscoat / Provisc Syringe

👉 WHAT TO DO RIGHT NOW:

Take the Viscoat syringe with a smooth 27-gauge cannula. Slide through the side port or main wound and inject the thick jelly directly over the back of the cornea. Next, inject cohesive Provisc to deepen the eye chamber.

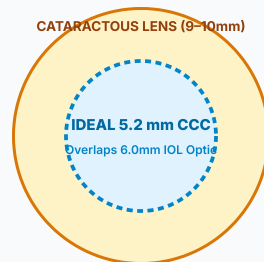
💡 **Why This Step Is Strictly Necessary:** The inner lining of the cornea is made of delicate "endothelial cells" that pump fluid out of the cornea to keep it crystal clear. Human endothelial cells DO NOT regenerate! If ultrasonic shockwaves, free radicals, or turbulent saline touch these cells, they die, turning the cornea permanently cloudy and white (bullous keratopathy), requiring a corneal transplant. The viscoelastic jelly acts like a protective shield of armor absorbing all ultrasound turbulence.

Arshinoff Soft-Shell Technique

1) Dispersive Viscoat coats and adheres tenaciously to endothelial cells; 2) Cohesive Provisc creates deep operating space and flattens the lens capsule.

⚠️ **Critical Hazard:** Over-pressurizing the eye causes zonular rupture or capsule blowout. Inadequate jelly coverage leaves corneal cells exposed to ultrasonic cell death.

CONTINUOUS CURVILINEAR CAPSULORHEXIS (CCC) VECTOR PHYSICS



BIOMECHANICAL TEAR VECTORS

✓ **SHEAR VECTOR (Safe - In-Plane Traction)**
Force directed tangentially along the circle.
Capsule tears smoothly without running away.

✗ **RADIAL STRETCH (Danger - Runaway Tear)**
Centrifugal vector pulls out toward zonules.
RESCUE: Pull flap flat 180° back to pupil center!

STEP 4

Continuous Curvilinear Capsulorhexis (CCC) — The Window in the Bag

Tool: Cystotome & Utrata Forceps

👉 WHAT TO DO RIGHT NOW:

1) Poke the center of the lens skin with the bent needle (cystotome) to raise a small flap; 2) Grasp the flap edge with Utrata forceps and drag in a continuous, smooth circle around the 5.2mm guide ring.

💡 **Why This Step Is Strictly Necessary:** The cataract is encapsulated in an elastic membrane 14 microns thin (thinner than plastic wrap). To suck out the cataract, you must open this bag. If you cut it with scissors or make jagged tears, the tears will zip down the sides to the back of the eye like a run in pantyhose. The whole cataract and lens will fall into the back of the eyeball (vitreous loss)! A continuous curvilinear circle has enormous tensile strength with zero stress points.

Target Dimensions	Diameter 5.0 to 5.5 mm (ideally 5.2 mm), centered precisely on the Purkinje optical visual axis.
Little's Rescue Technique	If the tear starts running out radially toward the edge, unfold the flap flat and pull 180° directly back toward the pupil center to redirect the tear.

⚠️ **Critical Hazard:** Radial tear run-out into equatorial zonules causes posterior capsule rupture and dropped nucleus.

STEP
5

Hydrodissection & Rotation Test — Loosening the Cataract with Water

Tool: Hydrodissection Cannula (BSS)

👉 WHAT TO DO RIGHT NOW:

Slip the flat cannula tip just under the edge of your circular capsular opening. Inject a gentle pulse of balanced salt solution (BSS). Watch the "golden fluid wave" roll across the back. Then gently tap and rotate the lens 360 degrees.

💡 **Why This Step Is Strictly Necessary:** The cataract is glued to the inside of its bag by thousands of sticky cellular fibers. If you try to chop or turn the cataract while it is glued down, you will rip the tiny suspension strings (zonules) holding the bag in place. Spraying water hydraulically cleaves all adhesions without mechanical stress.

⚠️ **Critical Hazard:** Injecting too fast or forcefully without decompressing the front chamber causes "capsular block syndrome"—the pressure blows the back of the bag wide open!

STEP
6

Phaco-Chop Nucleofractis & Aspiration — Pulverizing the Hard Core

Tool: Phaco Ultrasound Handpiece + Chopper

👉 WHAT TO DO RIGHT NOW:

Depress your foot pedal: Position 1 for water flow, Position 2 for vacuum, Position 3 for ultrasound vibration. Embed the phaco tip into the cataract core, bring the chopper in from the side, and pull together to split the cataract into 4 quadrants. Vacuum up each quarter.

💡 **Why This Step Is Strictly Necessary:** A hard cataract is up to 10mm wide and cannot fit through a tiny 2.4mm incision. The phaco needle vibrates 40,000 times a second, using microscopic cavitation bubbles to liquefy the hard rock so the vacuum can suck it out through the hollow center.

Foot Pedal Protocol	Pos 0: Off Pos 1: Continuous Irrigation Pos 2: Vacuum Aspiration Pos 3: Ultrasonic Cavitation Power
Safe Phaco Zone	Always perform chopping and emulsification at the iris plane in the center of the pupil. Never activate ultrasound within 1.5mm of the fragile back capsule.

⚠️ **Critical Hazard:** Post-occlusion surge: when a dense fragment suddenly clears the tip, vacuum spikes and can suck the posterior capsule into the needle, instantly tearing it!

STEP
7

Cortical Remnant Clearance (I/A) — Vacuuming the Soft Leftovers

Tool: Coaxial I/A Handpiece

👉 WHAT TO DO RIGHT NOW:

Switch to the Irrigation/Aspiration wand. Press pedal to Position 2 (suction). Move around the outer edges of the bag, occlude the fluffy cortical remnants, and pull them into the center to vacuum them away. Polish the back capsule until clean.

💡 **Why This Step Is Strictly Necessary:** Even after the rock-hard core is gone, fluffy cotton-like cellular fibers remain stuck to the bag walls. If left behind, these cells cause severe postoperative eye inflammation, sky-high eye pressure (glaucoma), and trigger cloudy scarring behind the new lens.

⚠️ **Critical Hazard:** If you accidentally suck on the transparent posterior capsule, fine radial wrinkles ("spider-web sign") appear! Release the foot pedal IMMEDIATELY to Position 0 to prevent tearing the bag.